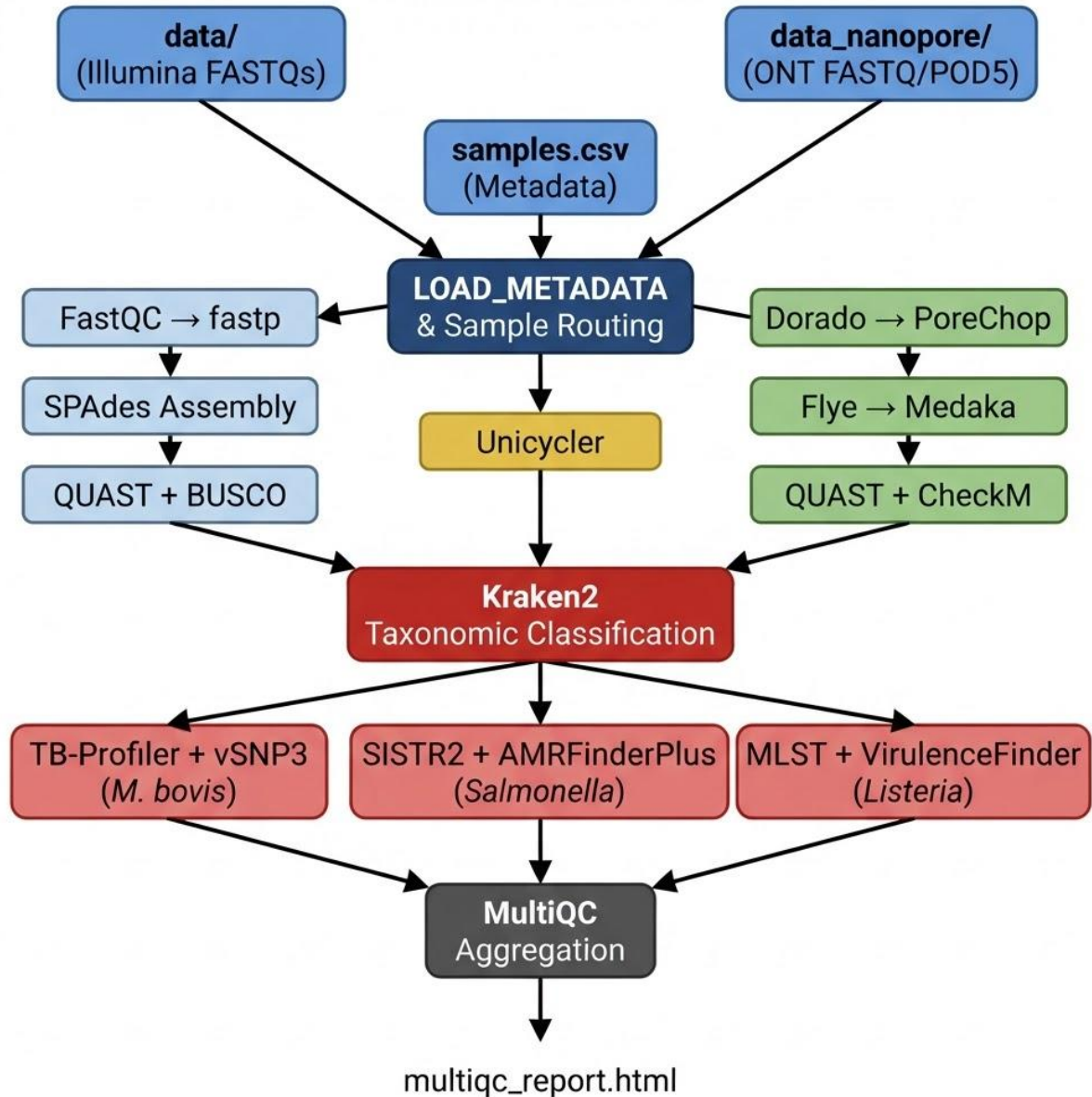


Hybrid QC, Assembly & Typing Pipeline

Illumina, Nanopore & Metagenomic Analysis

Workflow Diagram

The diagram below illustrates the pipeline architecture and data flow:



1 • Overview

This pipeline performs end-to-end quality control, genome assembly, and organism-specific typing for Illumina, Oxford Nanopore (ONT), and hybrid sequencing data. It is built in Nextflow DSL2, uses Docker and Conda for reproducibility, and can be driven by a metadata CSV manifest for explicit sample routing.

Pipeline Modes

- Standard Illumina: FastQC → fastp → SPAdes (--meta or --isolate) → QUAST/BUSCO → Kraken2 → Typing → MultiQC.
- Nanopore: Dorado (if raw POD5/FAST5) → PoreChop → Flye (--meta or isolate) → Medaka → QUAST/CheckM → Kraken2 → Typing → MultiQC.
- Hybrid: Pairs Illumina + ONT reads by Sample ID → Unicycler → QUAST/CheckM → Kraken2 → Typing → MultiQC.
- Metadata-Driven: Uses a samples.csv manifest to route samples to the correct branch automatically.

2 • Prerequisites & Installation

Core Software

- Nextflow (v25+): Required to run the pipeline.
- Docker: Used for most bioinformatics tools. Ensure your user is in the docker group.
- Conda: Used for vSNP3 (not containerized in Docker Hub).

Dorado Basecaller (GPU Required)

Dorado is Oxford Nanopore's basecaller for raw signal data (POD5/FAST5 files). A dedicated NVIDIA GPU is strongly recommended. Without a GPU, Dorado will run on CPU only, which is extremely slow — a single sample could take weeks to complete.

If your server does NOT have a GPU: Do not provide raw POD5/FAST5 files. Instead, basecall your data on a GPU-equipped workstation first (using MinKNOW or Dorado locally), and provide the resulting FASTQ files in the data_nanopore/ directory. The pipeline will automatically skip Dorado and proceed directly to PoreChop.

If your server HAS a GPU: Download the pre-compiled Linux binary from the Oxford Nanopore GitHub releases page and extract it to ~/dorado. Ensure CUDA drivers are installed. The pipeline will automatically detect POD5/FAST5 files and route them through Dorado.

Database Setup

- Kraken2: Download a prebuilt DB (e.g., k2_standard_8gb) and extract to ~/databases/kraken2.

- TB-Profiler: Run 'tb-profiler update_tbdb' inside the container to download the v6.x database to ~/databases/tbprofiler/tbdb.
- vSNP3: Download Mbovis_AF2122.fa and clone the vSNP_reference_options repo to ~/databases/vsnp/.
- VirulenceFinder: Clone the virulencefinder_db repo and run INSTALL.py to build the BLAST database in ~/databases/virulencefinder_db/.
- AMRFinderPlus: Run 'amrfinder -u' inside the container once to download the AMR database.

3 • Data Preparation & Metadata CSV

Place Illumina paired-end FASTQs in the data/ directory. Place Nanopore FASTQs or POD5/FAST5 files in the data_nanopore/ directory.

Important: If your server does not have a GPU, place pre-basecalled .fastq.gz files in data_nanopore/ — NOT raw .pod5 or .fast5 files. The pipeline will skip Dorado and process the FASTQs directly.

CSV File Structure

To use the metadata-driven workflow, create a samples.csv file in the project root. The order of columns does not matter, but the headers must match the expected names (case-insensitive). Save the file as CSV (UTF-8) format. See Appendix 1 for an example of a csv populated with permitted entry formats.

Required Headers

- sample_id: A unique identifier for each sample. This is the most critical field — it determines which workflow branch the sample is routed to.
- illumina_R1: The filename of the first Illumina read file (R1), located in the data/ directory.
- illumina_R2: The filename of the second Illumina read file (R2), located in the data/ directory.
- nanopore_file: The filename of the Nanopore read file (FASTQ, FAST5, or POD5), located in the data_nanopore/ directory.

Optional Headers

- Organism: Hint for the pipeline (mbovis, salmonella, listeria, virus, or leave blank to use Kraken2 auto-classification).
- Collection_Date, Source, Notes, Library_Type, Purpose, Contact: Metadata fields displayed in the MultiQC report.

How sample_id Routes Samples

The sample_id is the single most important field in the CSV. It determines which workflow branch a sample is routed to:

Illumina only: If `illumina_R1` and `illumina_R2` are filled and `nanopore_file` is empty, the sample runs through the Illumina branch (FastQC → fastp → SPAdes → QUAST/BUSCO).

Nanopore only: If `nanopore_file` is filled and `illumina_R1/illumina_R2` are empty, the sample runs through the Nanopore branch (Dorado/PoreChop → Flye → Medaka → QUAST/CheckM).

Hybrid: If BOTH Illumina and Nanopore fields are filled, the sample runs through the Hybrid branch (Unicycler → QUAST/CheckM). The `sample_id` ties the Illumina and Nanopore files together — they must share the same `sample_id` to be paired for hybrid assembly.

Note: The pipeline does NOT support Illumina single-end reads. The `illumina_R1` and `illumina_R2` fields must both be filled for Illumina processing. If you only have single-end Illumina reads, place the same file name in both R1 and R2 fields (the pipeline will handle it, though fastp may warn).

Example CSV Rows (see appendix 1)

Below are four example rows showing different sample types. In a real CSV, these would all be in the same file with the header row.

Example 1: Illumina Paired-End Only

This sample has Illumina R1 and R2 files but no Nanopore data. It will be routed to the Illumina branch.

```
MBWGS003,MBWGS003_R1.fastq.gz,MBWGS003_R2.fastq.gz,,mbovis,2026-06-20,CFIA-
Ottawa,isolate,Illumina,Routine Surveillance,asmith@lab.ca
```

Example 2: Illumina Single Read (Not Recommended)

The pipeline is designed for paired-end reads. If you only have single-end Illumina data, you must place the same filename in both R1 and R2 columns. The pipeline will run, but fastp may issue warnings.

```
SINGLE-01,SINGLE-01.fastq.gz,SINGLE-01.fastq.gz,,unknown,2026-08-15,Lab,clinical,Illumina
SE,Test,test@lab.ca
```

Example 3: Nanopore Only (Pre-basecalled FASTQ)

This sample has only Nanopore data (a pre-basecalled FASTQ file). The `illumina_R1` and `illumina_R2` fields are left empty. It will be routed to the Nanopore branch, skip Dorado, and proceed directly to PoreChop. Use this format if your server does not have a GPU.

```
ONT-Test01,,,ONT-Test01.fastq.gz,unknown,2026-08-26,Test,FASTQ test,Nanopore,Basecalling
Test,test@lab.ca
```

Example 3b: Nanopore Only (Raw POD5 — GPU Required)

This sample has raw POD5 signal data. The pipeline will route it through Dorado for basecalling. Only use this format if your server has a GPU installed.

ONT-Test02,,,sample_pod5.pod5,unknown,2026-08-26,Test,POD5 test,Nanopore,Dorado
Test,test@lab.ca

Example 4: Hybrid (Illumina + Nanopore)

This sample has BOTH Illumina paired-end reads AND a Nanopore FASTQ. Because all three file fields are filled, it will be routed to the Hybrid branch. The sample_id 'GTA-L201' ties the Illumina files and the Nanopore file together — Unicycler will use all of them to produce a hybrid assembly.

GTA-L201,GTA-L201_S1_L001_R1_001.fastq.gz,GTA-L201_S1_L001_R2_001.fastq.gz,GTA-L201.fastq.gz,listeria,2026-07-15,CFIA-Ottawa,metagenomic swab,Illumina+Nanopore,Outbreak Investigation,jdoe@lab.ca

Important Notes

- Leave fields empty with no spaces between commas if a file type is not applicable (e.g., ,, for no Nanopore file).
- Do NOT use the full file path — just the filename. The pipeline knows to look in data/ and data_nanopore/.
- The sample_id must be unique. If the same sample_id appears twice, only the first entry will be processed.
- File names are case-sensitive. Ensure they match exactly.
- Save the CSV as UTF-8 without BOM (Byte Order Mark). Excel may add a BOM by default; use 'Save As > CSV UTF-8' or a text editor.

If no GPU is available, provide .fastq.gz files, NOT .pod5 or .fast5 files. The pipeline will skip Dorado and process the FASTQs directly.

4 • Running the Pipeline

Standard Illumina Run

```
nextflow run . --kraken_db ~/databases/kraken2 --vsnp_ref ~/databases/vsnp/Mbovis_AF2122.fa --vsnp_options ~/databases/vsnp/reference_options/Mycobacterium_AF2122/Mbovis_define_filter.xlsx -tbprofiler_db ~/databases/tbprofiler/tbdb -resume
```

Nanopore Run (Metagenomic)

```
nextflow run . --run_nanopore true --nanopore_input 'data_nanopore/*' --assembly_mode meta --mem_spades '64 GB' --cpus_spades 16 -resume
```

Hybrid Assembly Run

```
nextflow run . --run_hybrid true --reads 'data/*_{r1,r2}.fastq.gz' --nanopore_input 'data_nanopore/*' --assembly_mode meta --kraken_db ~/databases/kraken2 -resume
```

Metadata-Driven Run (Recommended)

```
nextflow run . --samples_csv samples.csv --kraken_db ~/databases/kraken2 --vsnp_ref
~/databases/vsnp/Mbovis_AF2122.fa --vsnp_options
~/databases/vsnp/reference_options/Mycobacterium_AF2122/Mbovis_define_filter.xlsx --tbprofiler_db
~/databases/tbprofiler/tbdb --assembly_mode meta -resume
```

Testing Individual Modules

- Use the `--test` parameter to run specific modules in isolation. For example: `nextflow run . --test metadata --samples_csv samples.csv`

5 • Parameters Reference

Assembly Mode

- `--assembly_mode isolate` (default): For pure bacterial cultures. SPAdes uses `--isolate`, Flye omits `-meta`.
- `--assembly_mode meta`: For mixed/metagenomic samples. SPAdes uses `--meta`, Flye uses `--meta`.

Resource Overrides

- Global: `--global_cpus 8 --global_memory '32 GB'`
- SPAdes/Flye specific: `--cpus_spades 16 --mem_spades '64 GB'`
- BUSCO specific: `--cpus_busco 4 --mem_busco '8 GB'`

6 • Interpreting Results

The final MultiQC report is located at `results/multiqc/multiqc_report.html`. It contains:

- QC Reports: FastQC (raw/trimmed), fastp trimming stats.
- Retention & Checkpoints: Bar charts for read/base retention with 85% QC checkpoint line. PASS/FAIL for 95% basecalled, 85% QC, 80% assembly.
- Assembly Metrics: QUAST N50, BUSCO completeness, N50 bar chart, retention flow chart (Sankey), CheckM (for isolates).

- Typing Results: TB-Profiler (lineage, resistance), vSNP (phylogenetic group), SISTR2 (serovar), AMRFinderPlus (AMR genes), MLST (sequence type), VirulenceFinder (virulence genes).

7 • Troubleshooting

Permission denied when cleaning work/ directory

- Docker creates files as root. Use this command to delete locked directories: `docker run --rm -v "$(pwd)":/data alpine sh -c "rm -rf /data/YOUR_FILE_OR_FOLDER"`

Dorado runs extremely slowly or hangs

Dorado requires an NVIDIA GPU for efficient basecalling. Without a GPU, it runs on CPU only, which can take weeks for a single sample. If you do not have a GPU, do not provide raw POD5/FAST5 files. Basecall your data on a GPU-equipped workstation first, then provide the resulting .fastq.gz files in the data_nanopore/ directory. The pipeline will skip Dorado automatically.

Dorado Error: 'unknown modification variant: duplex'

- This occurs if the Dorado script uses 'sup,duplex'. In Dorado v0.7.3, duplex is not a modification variant. Use 'hac' or 'sup' directly in the modules/nanopore.nf script.

vSNP returns 'all_vcf' or 'NA' for all groups

- Ensure you are passing the --vsnp_options parameter pointing to the Mbovis_define_filter.xlsx file. Without this, vSNP step2 cannot assign phylogenetic groups.
- If all samples are assigned to 'all_vcf', it means there were too few samples to build a tree. This is expected for small test runs.

TB-Profiler shows 'NA' for all samples

- Ensure the --tbprofiler_db parameter points to the correct database path (~/databases/tbprofiler/tbdb). Also, the pipeline uses a Bash parser to extract lineage from the JSON; if the JSON structure changes, the parser in modules/typing.nf may need updating.

MultiQC reports 'No analysis results found'

- This happens when MultiQC doesn't receive enough valid report files. Ensure you pass --kraken_db so Kraken2 runs and produces a report. Also check that assemblies were not empty (0 bytes) — if Flye or SPAdes fail, downstream tools have nothing to process.

8 • Maintenance

The pipeline is modular. Process definitions live in the modules/ directory (e.g., qc.nf, assembly.nf, typing.nf). The main.nf file contains only parameters and workflow logic. To add a new tool, create a new process block in the relevant module file and include it at the top of main.nf.

- To update a tool version, change the container tag in `nextflow.config` and re-run with `-resume`. Nextflow will pull the new image and re-run the affected steps.

Appendix 1. Example of a metadata csv table.

sample_id	illumina_R1	illumina_R2	nanopore_file	Organism	Collection_Date	Source	Notes	Library_Type	Purpose	Contact
MBWGS003	MBWGS003_R1.fastq.gz	MBWGS003_R2.fastq.gz		mbovis	2026-06-20	CFIA-Ottawa	isolate	Illumina	Routine Surveillance	asmith@lab.ca
SINGLE-01	SINGLE-01.fastq.gz	SINGLE-01.fastq.gz		unknown	2026-08-15	Lab	clinical	Illumina SE	Test	test@lab.ca
ONT-Test01			ONT-Test01.fastq.gz	unknown	2026-08-26	Test	FASTQ test	Nanopore	Basecalling Test	test@lab.ca
ONT-Test02			sample_pod5.pod5	unknown	2026-08-26	Test	POD5 test	Nanopore	Dorado Test	test@lab.ca
GTA-L201	GTA-L201_S1_L001_R1_001.fastq.gz	GTA-L201_S1_L001_R2_001.fastq.gz	GTA-L201.fastq.gz	listeria	2026-07-15	CFIA-Ottawa	metagenomic swab	Illumina+Nanopore	Outbreak Investigation	jdoe@lab.ca

